Student's Name: .....

Arusha Regional Annual Examination December 2022

PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
ARUSHA REGION
FORM ONE ANNUAL EXAMINATION
PHYSICS

Time: 2:30 Hours **30th NOVEMBER 2022**

INSTRUCTIONS

1. This paper consist of three sections **A**, **B,** and **C** with a total of **ten (10)** questions
2. All writing must be in **blue** or **black** ink except drawing which must be in **pencil**
3. Write your **NAME** on top of every page of your answer sheets
4. Cellular phones, calculators and any unauthorized materials are not allowed in the examination room
5. Where necessary the following constants may be used:
   i. Acceleration due to gravity (g) = $10 m/s^2$
   ii. Pei (π) = 3.14
   iii. Density of water(δ) = $1 g/cm^3 or\ 1000 kg/m^3$

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SECTION A (15 Marks)

Answer **all** questions in the section

1. For each of the item (i)-(x), choose the correct answer from among the given alternatives and write its letter beside the item number in the answer sheet provided.

(i) When a screw gauge with a least count 0.01 mm is used to measure the diameter of a rod, the reading on the sleeve is found to be 1.6 cm and the reading on the thimble is found to be 48 divisions. What is the correct diameter of the rod if the zero error for the gauge is -0.003 cm ?

A. 0.651 cm B. 2.653 cm C. 1.651 cm D. 6.652 cm

(ii) The measurement of mass using a beam balance uses the principle of:

A. Conservation of matter B. Conservation of momentum
C. Gravitational pull of the Earth D. Moments

(iii) A relative density bottle has a mass of 18.50g when empty, 54.50g when filled with water and 38.30g when filled with second liquid. The density of the second liquid is

A. 1.8g/cm^3 B. 0.18g/cm^3 C. 0.055g/cm^3 D. 0.55g/cm^3.

(iv) A rectangular wooden block of density 0.8g/cm^3 has dimensions 0.5 m × 0.8 m × 6 m. What maximum pressure will it exert on the ground in N/m^2?

A. 48,000 B. 4800 C. 480 D. 48

(v) The angle of incidence is equal to the corresponding angle of reflection in:-

A. Diffuse reflection of light only. B. Regular reflection of light only
C. Both regular and diffuse reflection of light. D. Certain type of diffuse and regular reflection of light.

(vi) What is the mass of a man on the earth if his mass on the moon is 84 kg?

A. 8.4kg B. 14kg C. 84kg D. 504kg

(vii) A student gets electric shock and falls down unconscious in a Physics laboratory. Which of the following would you do first to help the victim?

A. Administer breathing exercise B. Call physics teacher C. Call other students D. Call medical doctor

(viii) An egg was immersed in a transparent liquid contained in a glass beaker. The egg was observed to float fully immersed in the middle of the liquid. What conclusion can you make from the observation?

A. The density of the egg is less than that of the liquid
B. The density of the egg is greater than that of the liquid
C. The egg and liquid have the same density
D. The upthrust on the egg is equal to the weight of the egg in air.

(ix) Which one of the following shows the energy changes in a hydroelectric plant (HEP)?

A. Heat $\longrightarrow$ Kinetic $\longrightarrow$ Electric
B. Potential $\longrightarrow$ Kinetic $\longrightarrow$ Electric
C. Electric $\longrightarrow$ Heat $\longrightarrow$ Kinetic
D. Kinetic $\longrightarrow$ Electric $\longrightarrow$ Potential

(x) The force which causes tear and wear between the machine parts is known as;

A. Friction B. Torsional C. Repulsive D. Magnetic

i	ii	iii	iv	v	vi	vii	viii	ix	x

2. Match the items in **List A** with those in **List B** by writing the letter of the corresponding response beside the item number in the answer sheet provided.

LIST A	**LIST B**
(i) Enables pond skater to walk on water surface. (ii) Absorption of water by a towel. (iii) Ratio of force applied and extension produced by it on a substance. (iv) Causes drop of mercury to be spherical in shape. (v) Tendency of a spring to recover its original shape after deformation.	A. Cohesive force B. Adhesive force C. Brownian motion D. Extension E. Surface tension F. Elasticity G. Elastic limit H. Osmosis I. Capillarity J. Kinetic theory K. Constant of a spring

LIST A	i	ii	iii	iv	v
LIST B					

SECTION B (70 marks)

Answer **all** questions in the section

3. (a) All form One students involved in performing physics experiment in the laboratory must be aware of safety measures. Give out four (4) measures to reduce risks during experiment.

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(b) Physicists are always after exact evidence. Write down six (6) critical and logical steps followed when carrying out a physics scientific investigation.

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Student's Name: ……

4. (a) Describe briefly the energy changes when the bob of the simple pendulum swings from one side to another.
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(b) A pendulum bob of mass 0.65 kg is pulled aside to a vertical height of 40 cm from the horizontal and released. Find:

(i) The maximum potential energy of the bob.

(ii) The maximum speed of the bob.

(c) Mention and state the principle applied by the pendulum experiment.
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(a) Why does a dull hypodermic needle hurt more than a sharp one?
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(d) Is there a net force on a dam due to atmospheric pressures? Explain your answer.
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(e) Some force is applied on the first cube (P_1) of side 4 m, and another force is applied on a cube (P_2) of side 2 m, where the magnitude of this force is double that of the previous force. Find the ratio of pressure on the first cube (P_1) to the pressure on the second cube (P_2).

5. (a) (i) With reference to Archimedes' principle, things submerged in a fluid can either sink, float or fly. Mention conditions that influence occurrence of each of these phenomena.
Sinking: ……
Floating: ……
Flying: ……

(ii) A body weighs 0.52N in air, 0.32N when totally immersed in water, and 0.36N when totally immersed in another liquid. Determine the density of the other liquid.

(b) (i) Briefly explain how buoyant force exerted by a fluid arise.
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(ii) A liquid mixture of water and alcohol has a density of 900kg/m^3. Find the mass of alcohol in one litre of the mixture. (One litre = 1000cm^3)

7. (a) A body dipped into a liquid experiences an upthrust. State two factors on which upthrust on the body depend.

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   (b) A metal solid cylinder tied to a thread is hanging from the hook of a spring balance. The cylinder is gradually immersed into the water contained in a jar. What changes do you expect in the readings of the spring balance? Explain your answer.

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   ……..……

   (c) Calculate the resulting force, if a steel ball of radius 6 cm is immersed in water.

8. (a) Four machines do the amounts of work listed in the table below. The time they take to do the work is also listed. Which machine develops the most power?

Machine	Work	Time
A	1000 joules	5 sec
B	1000 joules	10 sec
C	2000 joules	5 sec
D	2000 joules	10 sec

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   (b) What effect can a force have on a stationary body?

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9. (a) List two (2) characteristics of an image formed on a plane mirror.

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   (b) Explain the formation of multiple images on mirror inclined at 90°.

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   (c) Discuss the application of periscope.

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SECTION C (15 Marks)

10. (a) A form One student is in the physics laboratory. He is provided with a density bottle, sand, a digital balance, and water. He is required to determine the density of sand using the instruments provided above: show how will he determine the density of the sand?

    (b) In an experiment to determine the density of sand, Grace obtained and recorded the following results.

    Mass of the density bottle (M_1)= 200g

    Mass of density bottle and sand (M_2)= 490g

    Mass of density bottle, sand and water (M_3)= 550g

    Mass of density bottle and water (M_4)= 300g

    (i) What was the density of sand?

    (ii) Determine the relative density of sand.